

Anirban Majumdar

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Research Interests

Formal Methods, Model Checking, Automata Learning, Reinforcement Learning, (Partially Observable-) Markov Decision Processes, Reactive Synthesis, Parameterized Systems.

Academic Experience

Postdoctoral Researcher

December 2025–Present

Prof. R. Narasimhan Postdoctoral Fellow

February 2026–Present

Tata Institute of Fundamental Research, Mumbai

Mentor: Shibashis Guha

Research on reinforcement learning with timed specifications, Markov Decision Processes.

Independent Researcher

December 2024 – November 2025

Research on learning algorithms for one-counter automata, policy synthesis in Partially Observable Markov Decision Processes, reinforcement learning with timed specifications.

Postdoctoral Researcher

December 2021 – November 2024

Université Libre de Bruxelles, Belgium

Mentor: Jean-François Raskin

Research on learning algorithms for timed systems, strategy synthesis in Markov Decision Processes. Tool implementations for learning of event-recording automata, and Robbins' problem.

Education

Ph.D. in Computer Science

2018–2021

Université Paris-Saclay, France

Supervisors: Patricia Bouyer, Nathalie Bertrand

M.Sc. in Computer Science

2016–2018

Chennai Mathematical Institute, India

B.Sc. in Mathematics and Computer Science

2013–2016

Chennai Mathematical Institute, India

Publications

[\[DBLP\]](#) [\[Google Scholar\]](#)

1. About Time: Model-free Reinforcement Learning with Timed Reward Machines.

Accepted at IJCAI 2026, with Ritam Raha, Rajarshi Roy, David Parker and Marta Kwiatkowska.

2. Synthesizing POMDP Policies: Sampling Meets Model-checking via Learning.

Accepted at CAV 2026, with Debraj Chakraborty, Prince Mathew, Sayan Mukherjee and Jean-François Raskin.

3. Scalable Learning of One-Counter Automata via State-Merging Algorithms.

In *FSTTCS 2025*, with Shibashis Guha, Prince Mathew and A.V. Sreejith.

4. Learning Event-recording Automata Passively.

In *ATVA 2025*, with Sayan Mukherjee and Jean-François Raskin.

5. **Algorithms for Robbins' Problem Using Markov Decision Processes.**
In *Principles of Verification: Cycling the Probabilistic Landscape*, with Léonard Brice, Thomas F. Bruss and Jean-François Raskin.
6. **Greybox Learning of Languages Recognizable by Event-Recording Automata.**
In *ATVA 2024*, with Sayan Mukherjee, and Jean-François Raskin.
7. **Bi-objective Lexicographic Optimization in Markov Decision Processes with Related Objectives.**
In *ATVA 2023*, with Damien Busatto-Gaston, Debraj Chakraborty, Sayan Mukherjee, Guillermo A. Pérez and Jean-François Raskin.
8. **Reconfiguration and Message Losses in Parameterized Broadcast Networks.**
In *LMCS (2021)*, with Nathalie Bertrand and Patricia Bouyer.
9. **Playing with Repetitions in Data Words Using Energy Games.**
In *LMCS (2020)*, with Diego Figueira and M Praveen.
10. **Synthesizing Safe Coalition Strategies.**
In *FSTTCS 2020*, with Nathalie Bertrand and Patricia Bouyer.
11. **Computing the Width of Non-deterministic Automata.**
In *LMCS (2019)*, with Denis Kuperberg.
12. **Concurrent Parameterized Games.**
In *FSTTCS 2019*, with Nathalie Bertrand and Patricia Bouyer.
13. **Reconfiguration and Message Losses in Parameterized Broadcast Networks.**
In *CONCUR 2019*, with Nathalie Bertrand and Patricia Bouyer.
14. **Width of Non-deterministic Automata.**
In *Stacs 2018*, with Denis Kuperberg.
15. **Static and Dynamic Synthesis of Bengali and Devanagari Signatures.**
In *IEEE Transactions on Cybernetics (2018)*, with Moises Diaz, Sukalpa Chanda, Miguel A. Ferrer, Chayan Kr. Banerjee, Cristina Carmona-Duarte, Parikshit Acharya and Umapada Pal.
16. **Multiple Generation of Bengali Static Signatures.**
In *ICFHR 2016*, with Moises Diaz, Sukalpa Chanda, Miguel A. Ferrer, Chayan Kr. Banerjee, Cristina Carmona-Duarte, Parikshit Acharya and Umapada Pal.

Theses

Ph.D. Thesis: *Verification and Synthesis of Parameterized Concurrent Systems*
Supervisors: Patricia Bouyer-Decitre, Nathalie Bertrand

M.Sc. Thesis: *Playing with Repetitions in Data Words*
Supervisor: M. Praveen

Tools

- **tLsep** – Greybox Learning of ERA-recognizable languages.
Python implementation of an active learning algorithm for event-recording automata.
Source: github.com/mukherjee-sayan/ERA-greybox-learn
- **LEAP** – Passive Learning of ERA-recognizable languages.
Python implementation of a passive learning algorithm for event-recording automata.
Source: github.com/anirban11/leap
- **Robbins_MDP** – Solving Robbins’ Problem using MDPs.
Algorithmic exploration of optimal stopping problems using symbolic methods.
Source: github.com/anirban11/Robbins.MDP

Teaching Experience

- *Automata and Computability* Teaching Assistant, TIFR, 2026
- *INFO 412 - Formal verification of computer systems* Teaching Assistant, ULB, 2022–2024
- *INFO 410 - Embedded systems design* Teaching Assistant, ULB, 2022–2024

Programming skills

- **Languages:** Python, Haskell
- **Conceptual Knowledge:** Z3, TChecker, Uppaal, PRISM, XCos

Academic Internships

ENS Lyon, France	Summer 2017
Research on Good-for-Games Automata with Denis Kuperberg	
IMSc, Chennai	Summer 2016
Research on Bisimulation Equivalence for Pushdown Automata with Teodor Knapik	
ISI, Kolkata	Summer 2015
Research on Pattern Recognition with Umapada Pal	

Services

- **Organizing:** Local organizer of *CONFEST 2023*; volunteer for *FSTTCS 2024*.
- **Reviewing:** *FM 2026*, *MFCS 2025*, *ECAI 2025*, *AAMAS 2024*, *ICALP 2020*, *CONCUR 2020*.

Talks and Posters

- **Invited Talk:** *QuantFormal Workshop 2025*.
- **Conference Talks:** *FSTTCS 2025*, *ATVA 2024*, *FSTTCS 2020*, *FSTTCS 2019*, *CONCUR 2019*, *STACS 2018*.
- **Institute Seminars:** *Tata Institute of Fundamental Research*, *Chennai Mathematical Institute*, *Université Libre de Bruxelles*, *Inria Rennes*, *ENS Paris-Saclay*.
- **Poster:** *ATVA 2025*.